

Cosmic Shore

Repository state on 10 September 2026, measured against the Invite-Only Playtest checkpoint of 31 July. The build has grown far past what that plan described. One system in it does not run at all.

AUDITED	AGAINST	BRANCH	METHOD
10 Sep 2026	Checkpoint Rev 2 · 31 Jul	bleeding-edge 01685a89	Static repository audit

NOT COVERED
Off-machine work; editor runtime

PLAYABLE MODES 19 In the live arcade roster, all launchable. The checkpoint accounted for seven.	PROGRESSION SYSTEM Absent Present in source, instantiated by nothing. Fails open — nothing gates, nothing unlocks.	QA VERDICTS ON RECORD 1 Of 64 backlog items. 58 have never been run; the list is 145 PRs stale.	VESSEL ABILITY SLOTS 25/32 19 of 32 level-5 upgrades authored. Rhino is 1/4 and locks three shipped modes.	DEFINITION OF DONE 1/8 Clearly met today. Four are blocked on validation that has not started.
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01 ASSESSMENT

The checkpoint's central claim — *zero features left to build* — has held, and then some. What it did not anticipate is that the team would spend the six weeks building fifteen more game modes. The gap that matters is no longer feature work. It is that the one system the milestone's guided-progression story rests on is not in the build, and that essentially none of the new work has been validated.

Two things are worth separating, because they carry different risk. **Engineering delivery ran ahead of plan.** Workstream B, sized at 9.5 person-days, is largely landed as code and tooling: a scripted Windows IL2CPP pipeline, a SteamPipe uploader, offline fallback, crash reporting, four CI workflows. Workstream C's engineering half — settings, economy, dual-sink telemetry, consent — is done. The repository carries 73 build-gate scripts and 143 design documents; the discipline is real.

Validation did not run. Workstream D is 23 person-days of profiling, soak, regression, bug bash and a signed exit checklist, and there is no evidence in the repository that any of it began. The QA backlog was generated on 13 August against PRs up to #710; the repository is now at #855. It holds 64 items, of which 58 have never been run and 6 are partial, against exactly one submitted verdict in its entire history. That would be uncomfortable at the checkpoint's seven-mode scope. At nineteen modes it is the dominant risk to the milestone.

SCOPE NOTE ON THIS AUDIT

Everything below is verified against files in the repository. Workstreams A, E and most of F are off-machine — Steamworks paperwork, capsule art, a hosted privacy policy — and cannot be audited from here; they are marked as such rather than guessed at. The git clone available to this session covers 30 Aug – 10 Sep only, so no claim is made about what landed between 31 July and 30 August from commit history; mode and system inventory is taken from the shipped assets themselves.

The quest and progression system is not part of the build

`GameModeProgressionService` is a real, substantial system — 31 KB, cloud-persisted through `UGSDDataService.ProgressionRepo`, with unlock evaluation, intensity tiers, analytics hooks and a designer-tunable config. It is also instantiated by nothing. Its only host prefab sits in a folder named `Migration_Prefabs` (DELETE LATER) and is referenced by no scene, no prefab and no asset. No code path constructs it. At runtime `GameModeProgressionService.Instance` is null.

- 01 The service exists only on `Migration_Prefabs` (DELETE LATER)/`PlayerDataService.prefab`.
- 02 That prefab's guid appears in **zero** scenes, prefabs or assets across `Assets/`.
- 03 No `AddComponent`, `new GameObject` or `Resources.Load` creates it. Every consumer reads `.Instance` and null-guards.
- 04 Every null-guard **fails open**, so the failure is invisible: the game simply behaves as if everything is unlocked.
- 05 Independently, the authored `ProgressionConfig.asset` is **not assigned** to the service — its serialized block carries `questList` and `gameData` and no config field at all.

<code>ArcadeExploreView:255</code> <code>isLocked = false</code> for every card. The arcade is fully open.	<code>ArcadeGameConfigureModal:1110</code> Falls back to <code>game.MaxIntensity</code>. Every intensity is available.
<code>QuestTrackView</code> Null-guarded. The quest track renders nothing to progress.	<code>EndGameSequencer</code> No unlock is granted and no progression is written at end of match.

This reframes the checkpoint's own scoping. C10 describes “a data and configuration pass on a shipped system” and C11 “test and finalise the quest and progression system end to end” — eleven person-days that both assume the system is running. It is not. Definition of Done #4, “a fresh account completes the quest chain, with unlocks and progression persisting through Cloud Save,” cannot pass in the current build under any amount of dialogue authoring.

The second defect is the one that would bite *after* the first is fixed, and it is worth knowing before anyone wires the service up. `ProgressionConfig.asset` lists thirteen always-unlocked modes; because it is unassigned the service would fall back to its built-in default of `{ Maelstrom }` alone. Wiring the service in without also assigning that config would take the arcade from *fully open* to **fifteen of nineteen modes permanently locked** — leaving only Skim Race, Joust, Scurry and Maelstrom reachable, and hiding every mode built since the checkpoint. Three modes are missing from the authored list as well: Astro League 37, Brood Rush 38 and Scarab Scramble 43.

02.1 QUEST CONTENT, SEPARATELY

The designer's backlog is real but smaller than it looks, and it is not what blocks the milestone. The chain holds six quests covering four live modes — Crystal Capture, Hex Race, Joust and Wildlife Blitz — plus a Party Game entry explicitly flagged `IsPlaceholder: 1` and a Vessel Hangar unlock. Against nineteen playable modes, the chain describes about a fifth of the game.

Note also that `Assets/_Scripts/System/Quest/QuestSystem.cs` is a separate, older prototype and should not be mistaken for the shipping system: it carries hardcoded `TestQuest` / `TestQuest2` fields, keeps all state in memory with no persistence, has its reward grant commented out behind a `TODO`, and contains a loop that ignores its own iterator and always evaluates `TestQuest`. It is in no scene or prefab. It is dead code and a trap for the next person who greps for “quest”.

RECOMMENDATION — DECOUPLE THE BUILD FROM THE CHAIN

Make the quest chain a **guide, not a gate**, for this milestone.

`GameModeProgressionService` already carries `IsGameModeInQuestChain(mode)`, and the arcade view does not consult it. Having `IsGameModeUnlocked` return true for any mode outside the chain is a one-line change that gives the correct semantic — the chain leads a new player through four modes, and the rest of the game stays reachable — and it holds whether the chain has six quests or twenty.

That converts the designer's content from a critical-path blocker into an enhancement that can land at any time, and it lets the service be wired into the boot chain now so that Cloud Save persistence, unlock analytics and the quest track can actually be tested. The alternative — adding the three missing ids to `alwaysUnlockedModes` — works too, but leaves the same failure waiting for mode twenty.

03 WHAT SHIPPED SINCE THE CHECKPOINT

The checkpoint's “already shipped” table describes a minigame trio plus four bonus modes. That table is now substantially out of date, in the project's favour.

AREA	AT CHECKPOINT	NOW	NOTES
Game modes, live roster	7 (trio + 4 bonus)	19	All launchable; the gamelist scene gate passes across all seven rosters
Scenes in build settings	—	27	Bootstrap, Authentication, Menu_Main + 22 gameplay + 2 single-player
Vessel hulls	—	12	Scarab added beyond the eleven <code>VesselClassType</code> entries
Ability slots named	—	25/32	Dolphin, Sparrow, Squirrel, Urchin, Scarab complete
Level-5 upgrades authored	—	19/32	Manta, Rhino and Serpent carry none
First-party C# files	~1,100	1,878	
Build-gate scripts	—	73	<code>Tools/Build/*.py</code> , most runnable without Unity

Fifteen modes exist that the checkpoint's accounting does not contain: Astro League, Brood Rush, Rampage, Peel the Cage, Wildlife Liberation, Dog Fight, The Bends, Scarab Scramble, Salvo, Switchback, Hijack, Tollway, Headlong, Breakwater and Skein. Four of them — Switchback, Headlong, Breakwater, Skein — now share an extracted `GateRaceController` platform rather than each carrying their own copy, which is the right direction for a roster this size.

One coupling deserves attention. Astro League, Peel the Cage and Headlong all lock to the Rhino, whose elemental ability map has one of four slots designed and none of its four level-5 upgrades authored. Three shipped modes rest on a hull that is, by the fleet's own contract, unfinished. The Urchin has the mirror problem from the other side: its map is complete at 4/4 with all four upgrades, and it has no HUD prefab at all — `UrchinHUDVariant.prefab` does not exist, which leaves its HUD controller and view unreferenced. Both are design decisions, not defects, but they are exposed to players in the invite build.

04 WORKSTREAM REASSESSMENT

Each row is the checkpoint's own item, re-scored against what is actually in the repository. Off-machine items are marked rather than guessed.

ITEM	STATE	VERIFIED FINDING
A · Steam & Playtest A1–A7	OFF-MACHINE	Cannot be audited from the repository. <code>Docs/STEAM_BUSINESS_SETUP.md</code> is a thorough runbook — but it is written for Revision 1 : a paid release, the 30-day Steam Direct clock, and <code>beta / default</code> branches on one app. It contains no procedure for the Playtest child app, the Request Access queue or friend invites, so A4, A5 and A6 have no runbook behind them.
B1 · Windows build build profile	SHIPPED	<code>CosmicShoreBuildPipeline.cs</code> + <code>Tools/Build/build_windows.sh</code> drive <code>StandaloneWindows64</code> at IL2CPP/Release, stamp version and commit, and refuse a build whose first scene is not Bootstrap. Graphics APIs are explicitly D3D12 then D3D11, matching the device-gate fix. The <i>Build Profiles</i> folder holds a Linux profile only — documented as deliberate, since CI drives the script.
B2 · PC sanity pass 3.0 person-days	GAPS FOUND	Three mobile-era defaults were never re-pointed at PC: <code>defaultScreenWidth/Height</code> is 1024 × 768 , <code>resizableWindow</code> is 0 , and the Standalone bundle identifier is still <code>com.Froglet-Games.Tail-Glider</code> . The manual pass itself has no recorded result.
B3 · Offline launch	SHIPPED	<code>OfflineModeService</code> , <code>ReconnectService</code> and <code>LocalCloudDataCache</code> are present; the flow reaches the menu and states why.
B4 · SteamPipe both appIDs	SINGLE APP	<code>Tools/Steam/upload.sh</code> and the two VDF templates are ready and deliberately unarmed. They take one <code>STEAM_APPID / STEAM_DEPOTID</code> pair. Revision 2 requires the base app and the Playtest depot as separate targets.
B5 · Repeatable build B6 · Crash reporting	SHIPPED	<code>Docs/BUILD_AND_DELIVERY.md</code> carries the checklist and self-assesses each B item honestly. <code>CrashReportingService</code> sits behind the existing consent toggle.
B · CI runners	NOT RUNNING	Four workflows are authored — <code>unity-ci</code> , <code>bleeding-edge-guard</code> , <code>sync-build-branches</code> , <code>tag-internal-build</code> . The static half of the landing guard runs; the compile half needs a runner . No workflow builds a Windows player today.
C1 · Settings C5 · Economy	SHIPPED	Four-tab settings panel is live. <code>Docs/ECONOMY_TABLES.md</code> records the locked payout table (200 / 50 / 0) as applied in code and assets.
C4 · De-scope sweep 3.0 person-days	LARGELY OPEN	<code>StoreScreen</code> , <code>EpisodeScreen</code> , <code>PurchaseCard</code> and <code>PurchaseConfirmationModal</code> are all still live surfaces. The only “Coming Soon” treatment anywhere in the codebase is on quest cards. Partial credit: <code>MenuHubButton</code> does carry proper <i>Locked</i> and <i>Unavailable</i> states for Arena and Mission. The checkpoint calls this doubly load-bearing because the invite build is the first thing outsiders see — that reasoning still holds.
C7 · Instrumentation 5.0 person-days	PARTIAL	Both sinks exist (<code>PostHogAnalyticsSink</code> , <code>UgsAnalyticsSink</code>) behind a consent overlay, with a real event surface including <code>RecordModeUnlocked</code> , <code>RecordQuestCompleted</code> and the crystal economy. Wave cohorting appears nowhere in the instrumentation code — and the paid-EA gate is defined to read directly off those dashboards.

ITEM	STATE	VERIFIED FINDING
C9–C11 · Quest chain 15.0 person-days	MIS-SCOPED	See §02. The system is not in the build, so the “data and configuration pass on a shipped system” framing does not hold.
D · Stability gate 23.0 person-days	NOT STARTED	No evidence of D1–D6 in the repository. Docs/PERFORMANCE_OPTIMIZATION.md was last verified 8 July . No load-time target is published anywhere, which is D2's actual deliverable. The benchmark tool and its Load Time Insights tab are ready to produce one.
D · QA evidence	ONE VERDICT	64 backlog items: 58 never run , 6 partial, and a single submitted verdict — QA-BUILD-COMPILE PASS, 14 August. DEV_TASKS.md is empty. Critically, QA-BUILD-WINDOWS-PLAYER is a stated P0 and has never been run : no one has confirmed a Windows IL2CPP player reaches the main menu.
E · Marketing & waves 25.5 person-days	OFF-MACHINE	Not auditable here. The checkpoint's own HIGH risk — 13 days of capsule and trailer work on one contributor — is unchanged.
F · Legal 3.0 person-days	DRAFTED	Docs/Legal/ is present. Hosting the policy at a public URL and linking it are off-machine.

05 DOCUMENTATION DRIFT

Five documents will actively mislead someone picking up this work. They are cheap to fix and expensive to leave, because each one reads as authoritative.

DOCUMENT	DRIFT
Docs/STEAM_EA_INVESTOR_CHECKPOINT.html / .pdf	Is Revision 1 , 28 July, still describing a paid Early Access launch. The Playtest pivot exists nowhere in the repository — every runbook that cites it inherits the wrong destination.
Docs/QA/QA_BACKLOG.md generated 2026-08-13	Covers PRs #583–#710. The repository is at #855, so roughly 145 PRs of work — including nine game modes — are absent from the list QA is meant to work from.
Docs/UI_REDESIGN_TASKS.md T1, T3	Says T1 <i>in progress</i> and T3 <i>todo</i> . Both shipped: SafeAreaFitter exists, and the GameCanvas unification gate passes with the fork gone and 15 scenes on the CORE prefab.
Docs/PERFORMANCE_OPTIMIZATION.md verified 2026-07-08	Two months and roughly fifteen game modes stale. Its backlog is the input to D1 and D3.
Checkpoint Rev 2, item B2 “Wwise audio init”	The project's audio middleware is FMOD . Assets/Wwise/ has zero first-party references and is inert. A PC sanity pass written against Wwise would test nothing.

06 WHAT TO DO, IN ORDER

Ordered by what unblocks the most. The first three are days of work, not weeks, and two of them are independent of the designer.

01 Decide the progression posture, then wire the service in

BLOCKER

Take the guide-not-gate decision in §02, wire `GameModeProgressionService` into the boot chain, and assign `ProgressionConfig.asset`. Until this lands, Cloud Save persistence, unlock analytics and the quest track are all untestable, and Definition of Done #4 is unreachable. Do not wire the service without the config — that trades an open arcade for fifteen permanently locked modes.

02 Prove a Windows player reaches the menu

P0

`QA-BUILD-WINDOWS-PLAYER` is a stated P0 that has never been run. Every prior defect in that chain — the linker failure, the pause-menu crash — was invisible in the Editor. Nothing else in Workstream D means anything until this passes, and it also unblocks B7's overlay check.

03 Regenerate the QA backlog and start the daily pass

CRITICAL PATH

Run `/qa-backlog` so the list covers PRs #711–#855 and the nine modes added since. C8's daily all-modes session now means nineteen modes, not seven — it should start immediately rather than waiting for the bug bash, because D5 needs a seed corpus that does not currently exist.

04 Re-scope D2 against the real matrix

RE-SCOPE

Load time was sized at 8.0 person-days against roughly seven modes. The matrix is now **19 modes × 4 intensities = 76 combinations**. Publish the load-time target first — it is D2's actual deliverable and it does not exist yet — then measure. The benchmark tool's Load Time Insights tab already produces the attribution.

05 Finish the de-scope sweep

3.0 DAYS

Give the store, episode and purchase surfaces the deliberate coming-soon treatment C4 specifies. `MenuHubButton`'s Locked and Unavailable states are the pattern to extend — the mechanism exists, it simply has not been applied to the commerce surfaces.

06 Write the Playtest half of the Steam runbook

WORKSTREAM A/B

A4, A5 and A6 have no procedure behind them, and the existing runbook points at a paid release. Add the child app, Request Access queue and friend-invite configuration, and extend `Tools/Steam/upload.sh` to take a second appID/depot pair.

07 Add wave cohorting to instrumentation

GATE INPUT

The paid-EA gate is defined to read retention and stability cohorted by invite wave. No cohort dimension exists in the analytics code today, so the gate has no data source. This is small, and it is on the critical path to the milestone *after* this one.

08 Close the PC platform defaults and the doc drift

LOW COST

Set the default resolution to 1920 × 1080, enable `resizableWindow`, re-point the Standalone bundle identifier off *Tail-Glider*, and correct the five documents in §05 — starting with publishing Revision 2 into the repository so the runbooks stop describing a paid launch.

07 DEFINITION OF DONE, RE-SCORED

The checkpoint requires all eight to hold simultaneously. One does today.

1	Steam page live with Playtest signup, review passed. Off-machine, and the Playtest half has no runbook.	AT RISK
2	Playtest build approved, Waves 0 and 1 granted, friend invites on. Blocked on #1 and on a proven Windows player.	BLOCKED
3	Fresh install reaches a completed four-player Tournament, zero P0/P1. Unproven — the P0 player-build test has never been run and there is no defect corpus to score against.	UNPROVEN
4	Fresh account completes the quest chain, unlocks persist through Cloud Save. Cannot pass: the progression service is not instantiated in any scene.	CANNOT PASS
5	Cold boot to playable meets the load-time target in every mode at every intensity. No target published; the matrix is now 76 combinations.	NO TARGET

6	Crash reporting and funnel telemetry in dashboards, cohorted by wave. Crash reporting and both analytics sinks ship; wave cohorting does not exist.	PARTIAL
7	Zero public review surface. Structural — a property of Steam Playtest configuration, not of the build.	MET
8	Paid-EA gate criteria published and exit checklist signed. Criteria are published in Revision 2; the checklist cannot be signed without Workstream D evidence.	HALF

The honest summary is that the milestone is not gated on building anything. It is gated on **one wiring decision, one player build, and the validation pass that was scoped in July and has not begun** — now against a game roughly three times the size the plan assumed. The engineering that the checkpoint said was finished really is finished, and then some; what is missing is the evidence that it works.